Introduction to Enrollment Forecasts

Portland Public Schools

January 22, 2015



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Population Research Center
College of Urban and Public Affairs

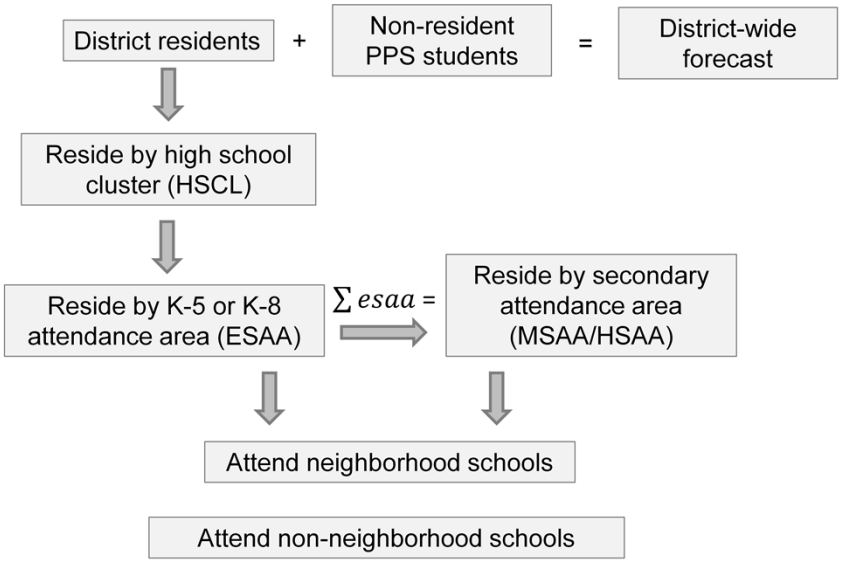
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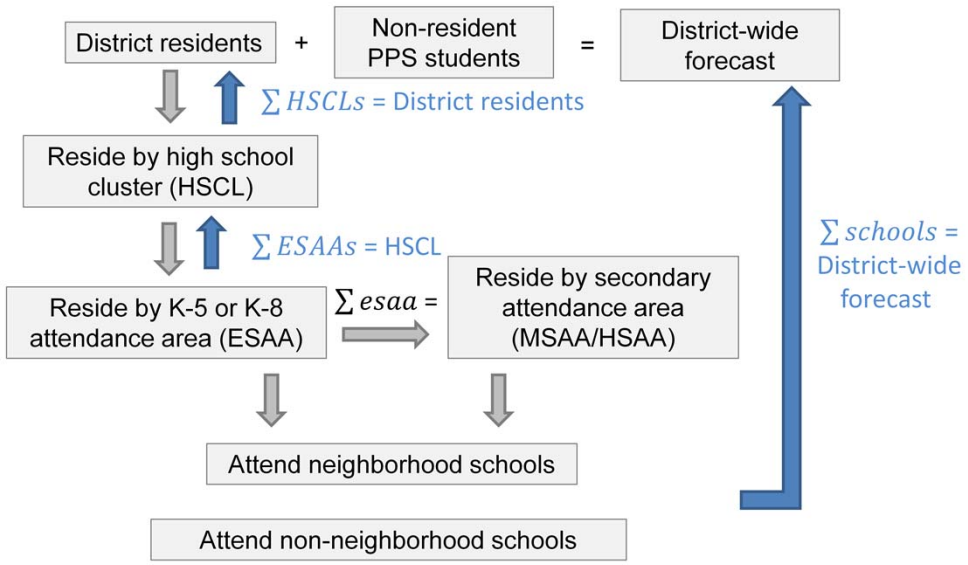
Outline

- Forecast process and methodology
- Population and housing trends
- Enrollment trends
- Forecast accuracy

Forecast Methodology: Top-Down



Forecast Methodology: Controlling



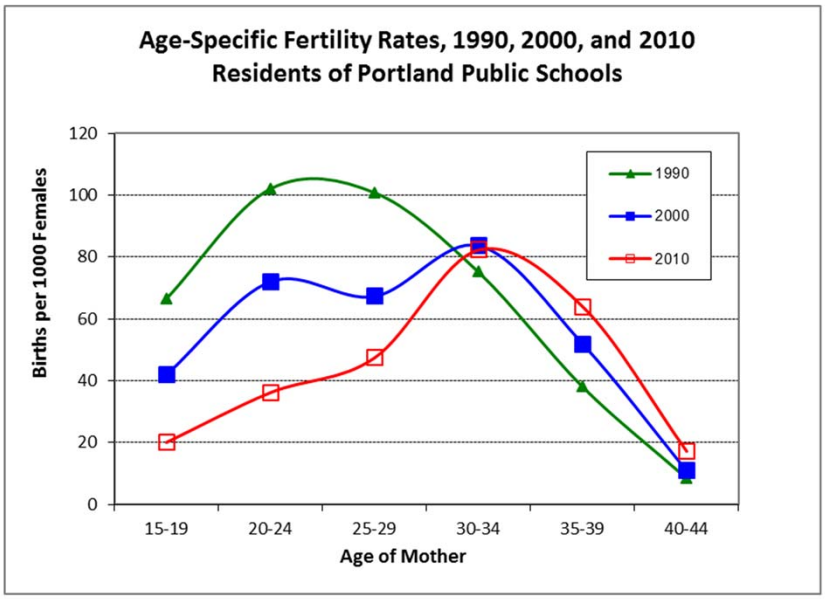
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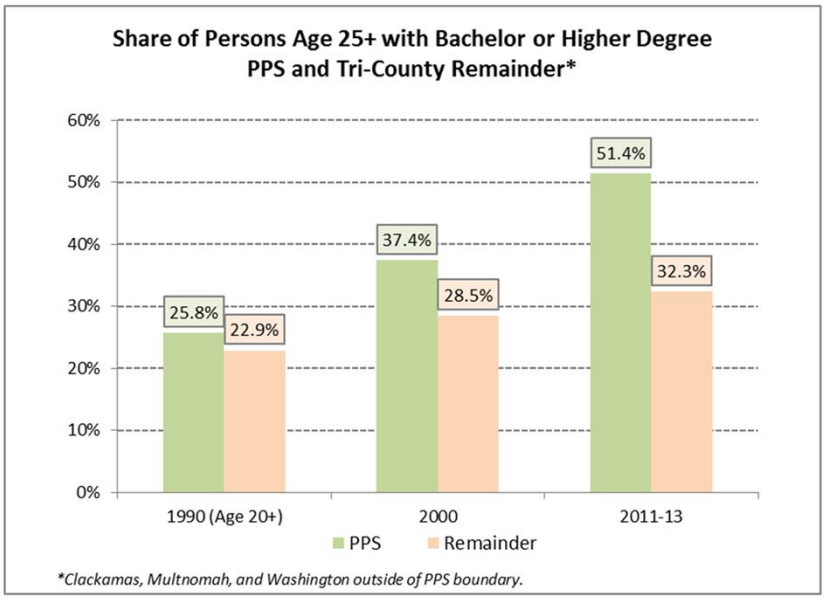
District-wide Forecast

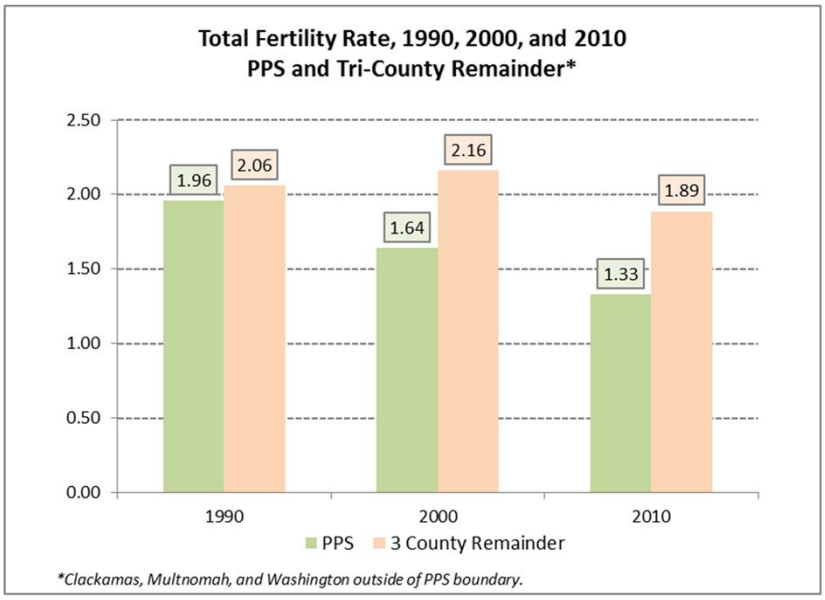
District residents: cohort-component model

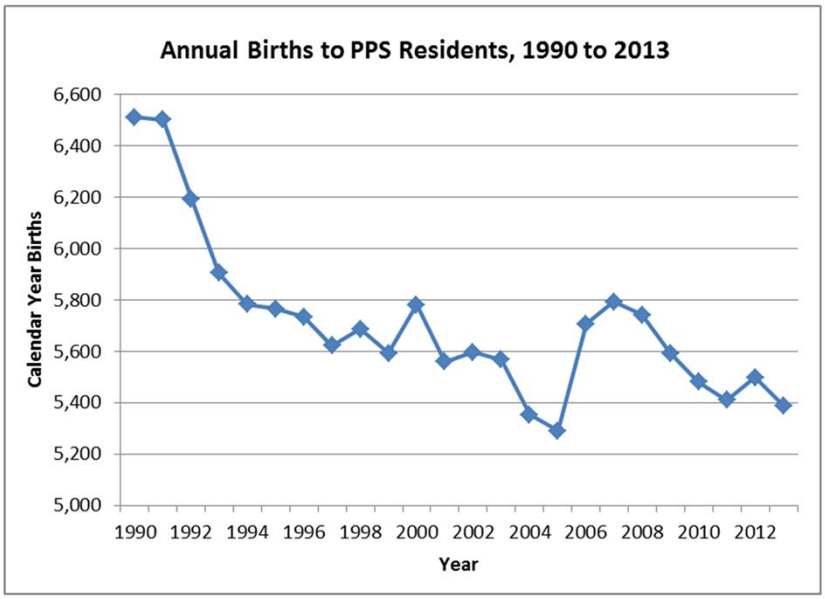
- Historic enrollment
- Fertility rates, births
- "Capture rates"
- "Baseline" grade progression rates (GPRs)
- Census data, migration assumptions
- Population forecast: single years 0-19, five year age groups

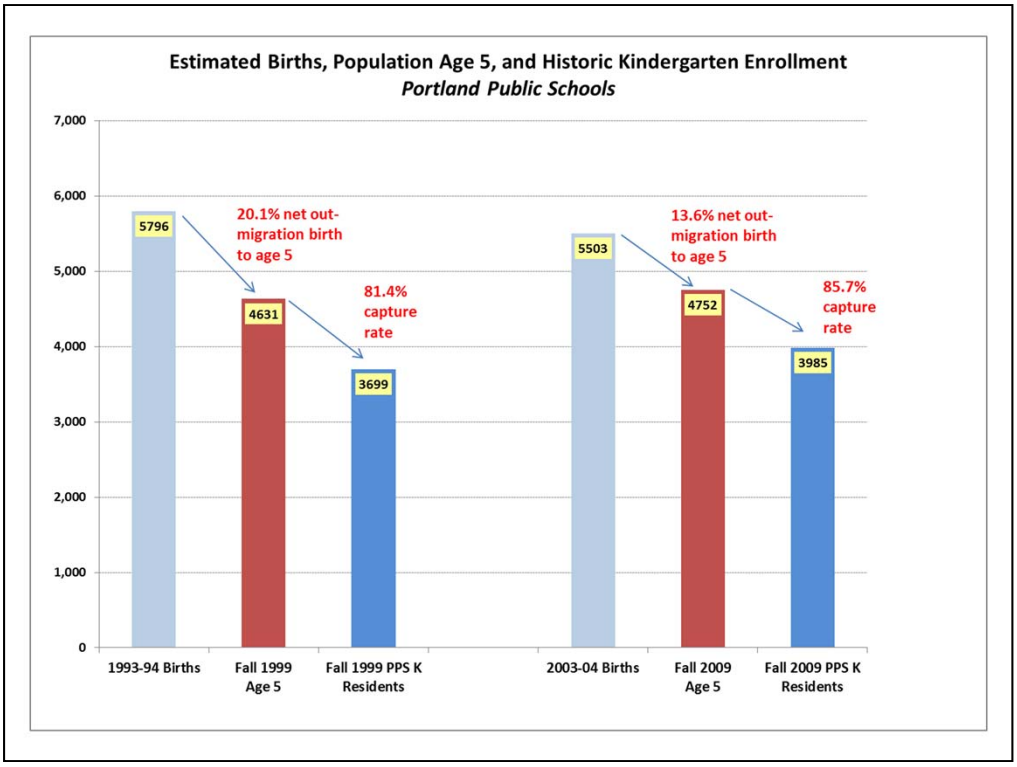
Non-residents: historic trends and GPRs


Age-Specific Fertility Rates, 1990, 2000, and 2010
Residents of Portland Public Schools
Births per 1000 Females
120
100
80
60
40
20
0
15-19
20-24
25-29
30-34
35-39
40-44
Age of Mother
1990
2000
2010


Share of Persons Age 25+ with Bachelor or Higher Degree
PPS and Tri-County Remainder*
60%
50%
40%
30%
20%
10%
0%
25.8%
22.9%
37.4%
28.5%
51.4%
32.3%
1990 (Age 20+)
2000
2011-13
PPS
Remainder
*Clackamas, Multnomah, and Washington outside of PPS boundary.


Total Fertility Rate, 1990, 2000, and 2010
PPS and Tri-County Remainder*
2.50
2.00
1.50
1.00
0.50
0.00
1.96
2.06
1.64
2.16
1.33
1.89
1990
2000
2010
PPS
3 County Remainder
*Clackamas, Multnomah, and Washington outside of PPS boundary.


Annual Births to PPS Residents, 1990 to 2013
Calendar Year Births
6,600
6,400
6,200
6,000
5,800
5,600
5,400
5,200
5,000
1990 1992 1994 1996 1998 2000 2002 2004 2006 2008 2010 2012
Year


Estimated Births, Population Age 5, and Historic Kindergarten Enrollment
Portland Public Schools
7,000
6,000
5,000
4,000
3,000
2,000
1,000
0
5796
4631
3699
5503
4752
3985
20.1% net out-migration birth to age 5
81.4% capture rate
13.6% net out-migration birth to age 5
85.7% capture rate
1993-94 Births
Fall 1999 Age 5
Fall 1999 PPS K Residents
2003-04 Births
Fall 2009 Age 5
Fall 2009 PPS K Residents

Grade Progression Rates (GPRs)

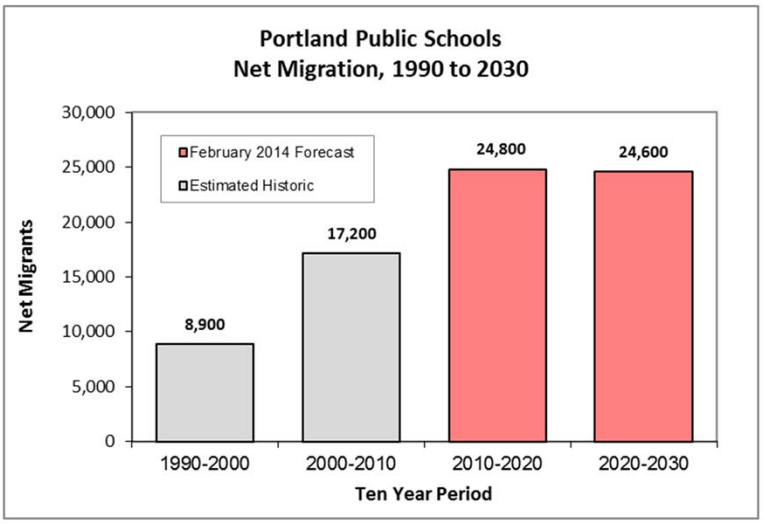
$year_y$ enrollment in $grade_x$ ÷ $year_{y-0}$ enrollment in $grade_{x-0}$

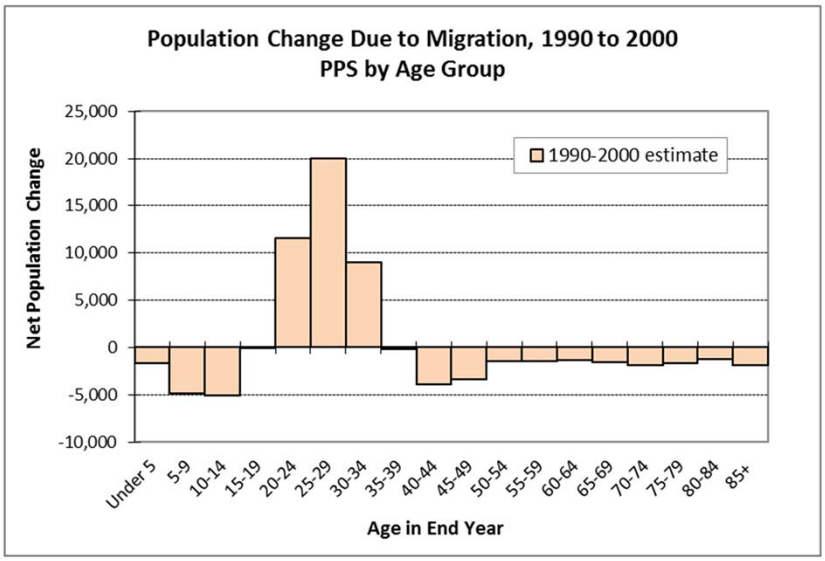
	2008-13 5 yr avg	forecast baseline	2009-14 5 yr avg	forecast baseline
K-1	1.011		1.011	
1-2	0.980	1.000	0.980	1.000
2-3	0.987	1.000	0.986	1.000
3-4	0.990	1.000	0.994	1.000
4-5	0.983	1.000	0.985	1.000
5-6	0.972	0.985	0.968	0.980
6-7	0.988	1.000	0.981	1.000
7-8	0.983	1.000	0.984	1.000
8-9	0.970	0.980	0.958	0.975
9-10	0.999	1.010	1.003	1.010
10-11	0.952	0.960	0.955	0.970
11-12	1.103	1.110	1.123	1.140
2-5/1-4	0.985		0.986	

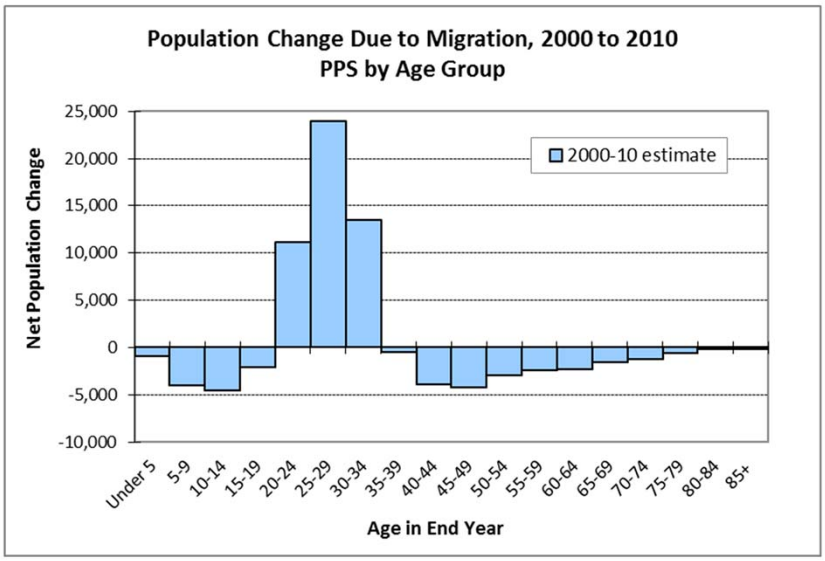
Historic averages incorporate migration, school choice, dropouts, and retention.

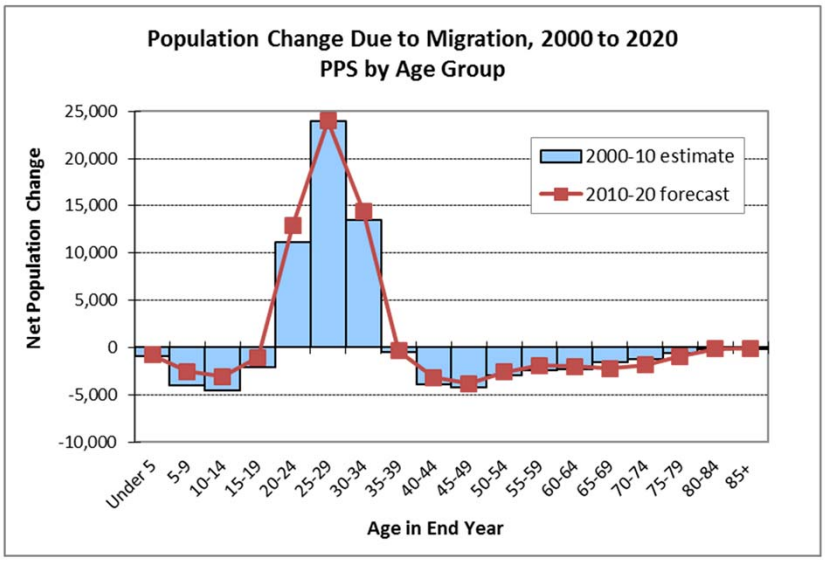
Baseline rates are future assumptions for GPRs excluding the migration component.

Grade progression rates (GPRs) measure the ratio of enrollment in a specific grade to the previous year's enrollment in the previous grade. Some forecasters call them "cohort survival rates." They incorporate migration, school choice, dropouts, and retention. The "5 yr avg" columns are averages of the most recent 5 years of grade progression rates, whereas the "baseline" columns contain the future assumptions for GPRs excluding the migration component. There is no baseline K-1 GPR because capture rates are used in the model for both kindergarten and 1st grade.


Portland Public Schools
Net Migration, 1990 to 2030
30,000
25,000
20,000
15,000
10,000
5,000
0
Net Migrants
February 2014 Forecast
Estimated Historic
8,900
17,200
24,800
24,600
1990-2000
2000-2010
2010-2020
2020-2030
Ten Year Period

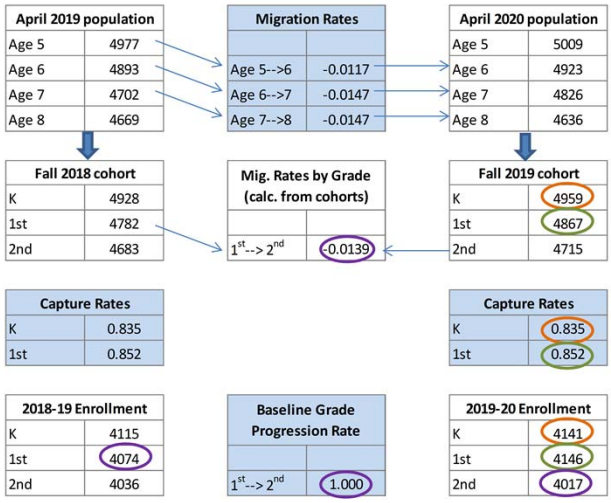

Population Change Due to Migration, 1990 to 2000
PPS by Age Group
Net Population Change
25,000
20,000
15,000
10,000
5,000
0
-5,000
-10,000
1990-2000 estimate
Under 5
5-9
10-14
15-19
20-24
25-29
30-34
35-39
40-44
45-49
50-54
55-59
60-64
65-69
70-74
75-79
80-84
85+
Age in End Year


Population Change Due to Migration, 2000 to 2010
PPS by Age Group
Net Population Change
25,000
20,000
15,000
10,000
5,000
0
-5,000
-10,000
2000-10 estimate
Under 5
5-9
10-14
15-19
20-24
25-29
30-34
35-39
40-44
45-49
50-54
55-59
60-64
65-69
70-74
75-79
80-84
85+
Age in End Year


Population Change Due to Migration, 2000 to 2020
PPS by Age Group
Net Population Change
25,000
20,000
15,000
10,000
5,000
0
-5,000
-10,000
2000-10 estimate
2010-20 forecast
Under 5
5-9
10-14
15-19
20-24
25-29
30-34
35-39
40-44
45-49
50-54
55-59
60-64
65-69
70-74
75-79
80-84
85+
Age in End Year

Example from PPS Resident Model

Derivation of 2019-20 forecast for K, 1st, and 2nd grade (prepared February 2014)



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District Forecast Including Non-Residents

2019-20 PPS Resident Enrollment	
K	4,141
1st	4,146
2nd	4,017

+

2019-20 PPS Non-Resident Enrollment	
K	90
1st	99
2nd	93

=

2019-20 Total Enrollment (Medium Scenario)	
K	4,231
1st	4,245
2nd	4,110

Forecasts: Residents by High School Cluster (HSCL)

- Specific residential developments or neighborhood characteristics explicitly or implicitly incorporated
- Kindergarten based on births and historic trends; control sum of 8 HSCLs to district-wide resident forecasts
- Subsequent grades 1st to 12th based on GPRs; control sum of 8 HSCLs to district-wide resident forecasts

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Forecasts: Residents by Attendance Area (AA)

- Specific residential developments or neighborhood characteristics explicitly or implicitly incorporated
- Kindergarten based on historic trends; births considered but relationship weaker than in HSCL or district-wide models; control sum of AAs to HSCL forecast
- Subsequent grades 1st to 12th based on GPRs; control sum of AAs to HSCL forecast

Forecasts: Neighborhood Schools

Residents

- "Capture rate" for incoming grade (K, 6th, or 9th) = expected share of AA residents
- GPRs for subsequent grades
- Check for consistency (stability or trend) in school's share of AA residents

Non-residents

- Incoming grade based on history, trend, or information from PPS (e.g. focus program, capacity constraint, etc.)
- GPRs for subsequent grades

Total = residents + non-residents

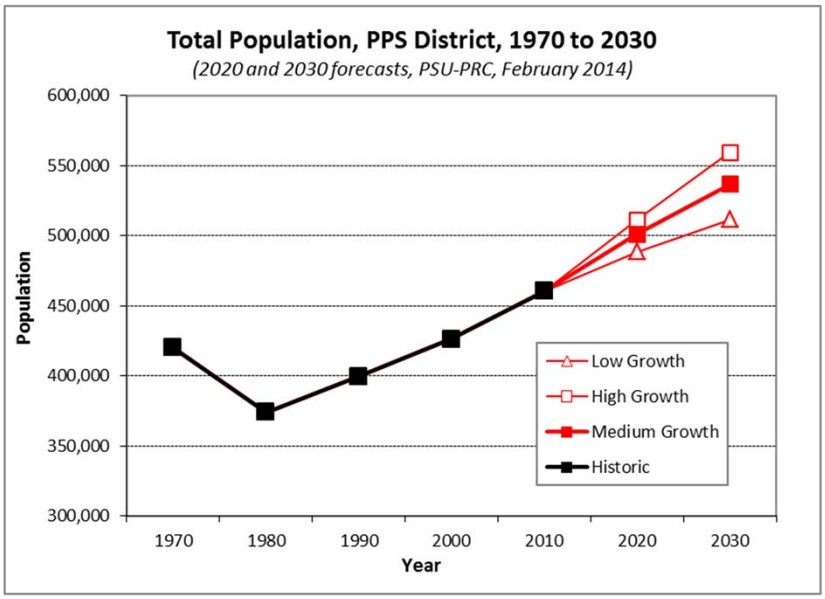
Forecasts: Non-Neighborhood Schools

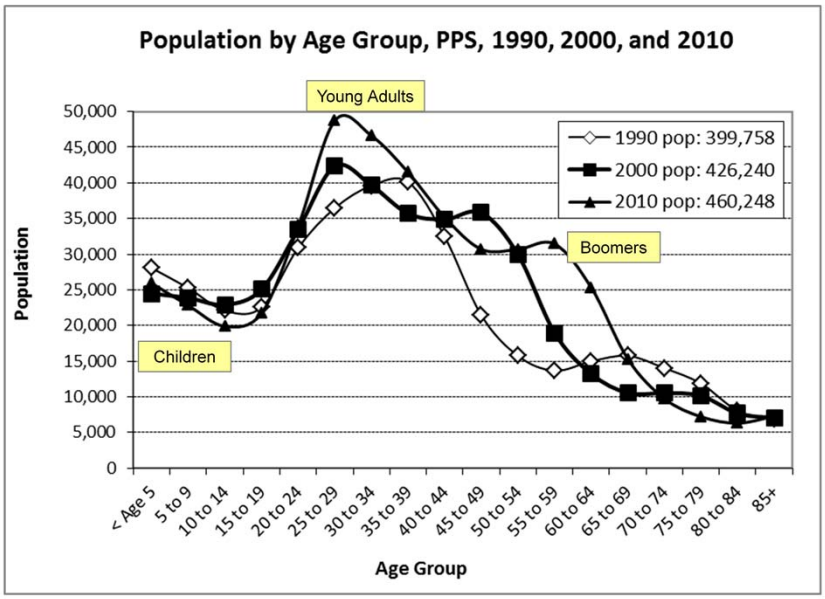
- Specific forecasts prepared for 8 district-wide programs
- Incoming grade based on history or information from PPS
- GPRs for subsequent grades
- Constrained by capacity or policy, with relatively stable enrollments
- "Other schools and programs": residual of district-wide forecast minus neighborhood and non-neighborhood school forecasts, with a GPR model to check for reasonableness

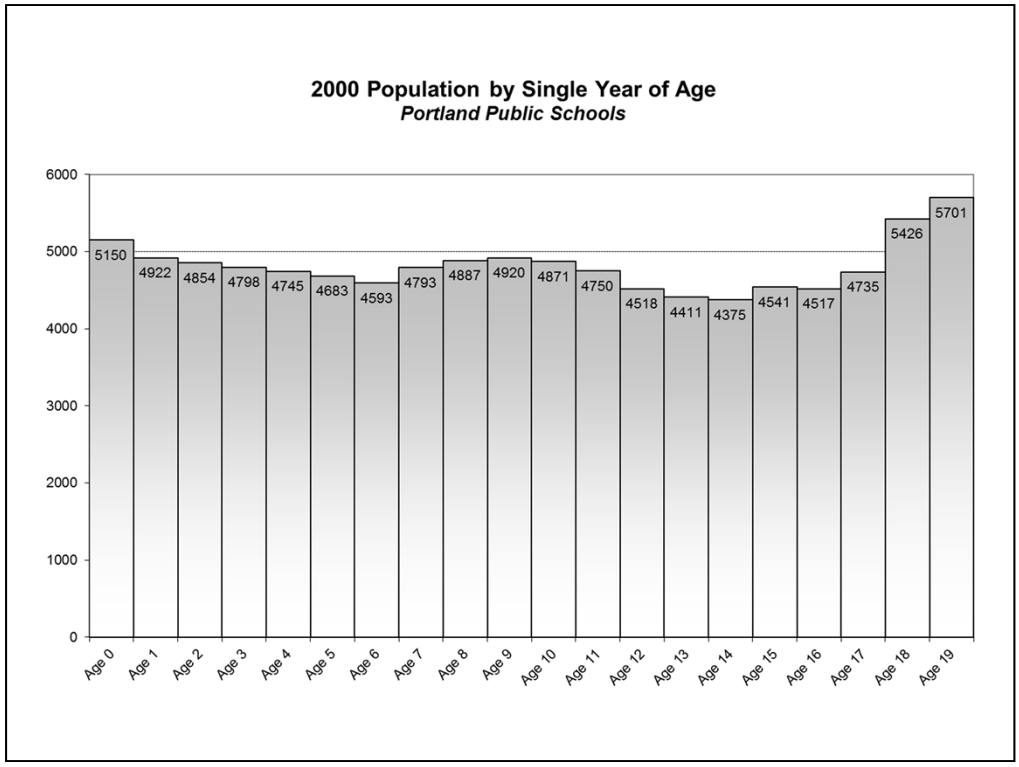
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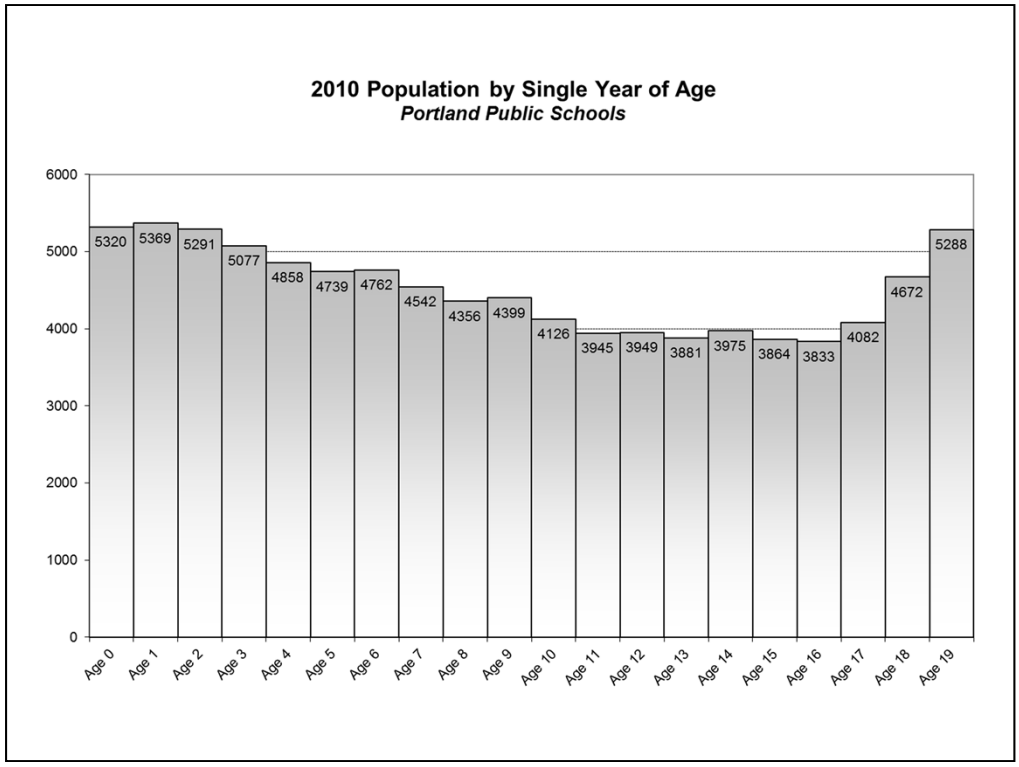
Population and Housing Trends

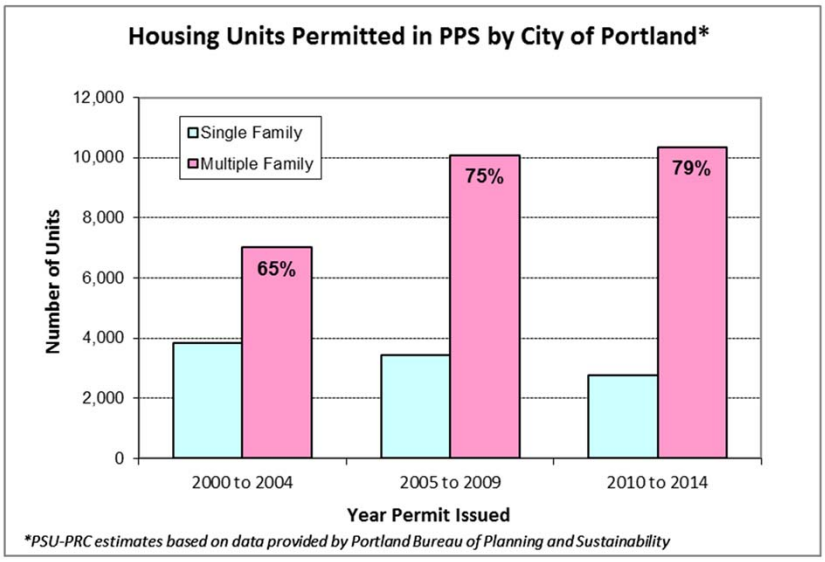
- Total Population
- Population by age
- Housing growth
- Student generation rates by housing type

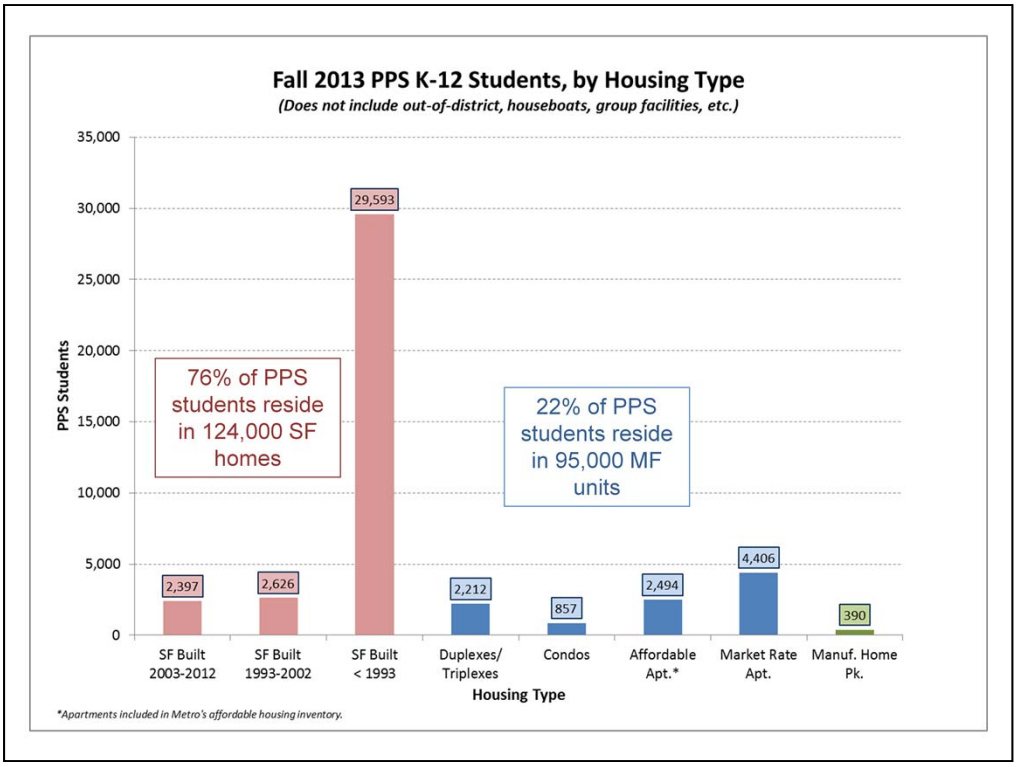

Total Population, PPS District, 1970 to 2030
(2020 and 2030 forecasts, PSU-PRC, February 2014)
600,000
550,000
500,000
450,000
400,000
350,000
300,000
Population
1970
1980
1990
2000
2010
2020
2030
Year
Low Growth
High Growth
Medium Growth
Historic


Population by Age Group, PPS, 1990, 2000, and 2010
Young Adults
Boomers
Children
1990 pop: 399,758
2000 pop: 426,240
2010 pop: 460,248
Population
50,000
45,000
40,000
35,000
30,000
25,000
20,000
15,000
10,000
5,000
0
< Age 5
5 to 9
10 to 14
15 to 19
20 to 24
25 to 29
30 to 34
35 to 39
40 to 44
45 to 49
50 to 54
55 to 59
60 to 64
65 to 69
70 to 74
75 to 79
80 to 84
85+
Age Group


2000 Population by Single Year of Age
Portland Public Schools
6000
5000
4000
3000
2000
1000
0
5150
4922
4854
4798
4745
4683
4593
4793
4887
4920
4871
4750
4518
4411
4375
4541
4517
4735
5426
5701
Age 0
Age 1
Age 2
Age 3
Age 4
Age 5
Age 6
Age 7
Age 8
Age 9
Age 10
Age 11
Age 12
Age 13
Age 14
Age 15
Age 16
Age 17
Age 18
Age 19


2010 Population by Single Year of Age
Portland Public Schools
6000
5000
4000
3000
2000
1000
0
5320
5369
5291
5077
4858
4739
4762
4542
4356
4399
4126
3945
3949
3881
3975
3864
3833
4082
4672
5288
Age 0
Age 1
Age 2
Age 3
Age 4
Age 5
Age 6
Age 7
Age 8
Age 9
Age 10
Age 11
Age 12
Age 13
Age 14
Age 15
Age 16
Age 17
Age 18
Age 19


Housing Units Permitted in PPS by City of Portland*
12,000
10,000
8,000
6,000
4,000
2,000
0
Number of Units
Single Family
Multiple Family
65%
75%
79%
2000 to 2004
2005 to 2009
2010 to 2014
Year Permit Issued
*PSU-PRC estimates based on data provided by Portland Bureau of Planning and Sustainability


Fall 2013 PPS K-12 Students, by Housing Type
(Does not include out-of-district, houseboats, group facilities, etc.)
PPS Students
35,000
30,000
25,000
20,000
15,000
10,000
5,000
0
76% of PPS students reside in 124,000 SF homes
22% of PPS students reside in 95,000 MF units
2,397
2,626
29,593
2,212
857
2,494
4,406
390
SF Built 2003-2012
SF Built 1993-2002
SF Built < 1993
Duplexes/ Triplexes
Condos
Affordable Apt.*
Market Rate Apt.
Manuf. Home Pk.
Housing Type
*Apartments included in Metro's affordable housing inventory.

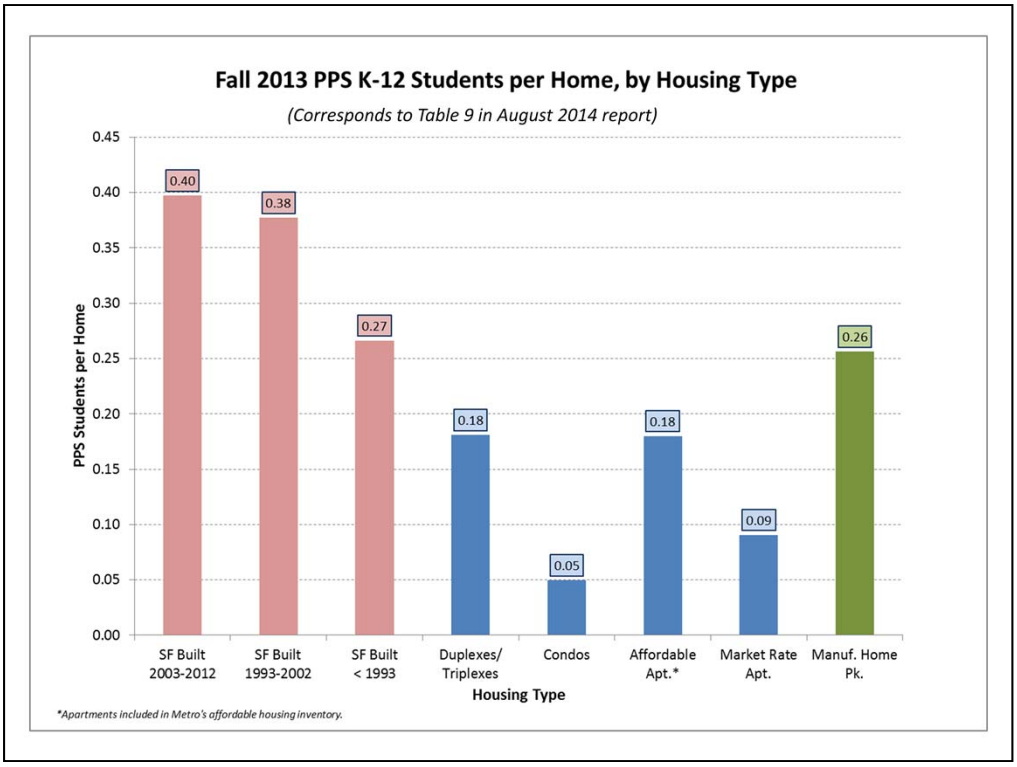

Fall 2013 PPS K-12 Students per Home, by Housing Type
(Corresponds to Table 9 in August 2014 report)
PPS Students per Home
0.45
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
0.40
0.38
0.27
0.18
0.05
0.18
0.09
0.26
SF Built 2003-2012
SF Built 1993-2002
SF Built < 1993
Duplexes/ Triplexes
Condos
Affordable Apt.*
Market Rate Apt.
Manuf. Home Pk.
Housing Type
*Apartments included in Metro's affordable housing inventory.

Table 10. Housing Types

Excerpt from Comprehensive Plan Update, Growth Scenarios Report, City of Portland, May 2013. Full document at http://www.portlandoregon.gov/bps/article/449310.
Estimated K-12 Student Generation Rates for PPS added by PSU-PRC (red text)

SINGLE FAMILY RESIDENCES



Detached House
A one- to three-story detached, single family dwelling on its own lot. Typically, lot size is more than 5,000 square feet.

<10 years old 0.40
10-19 years old 0.38
20+ years old 0.27



Small Lot Single Family Residence
A one- to three-story detached, single family dwelling on its own lot, but a smaller (2500 sq foot) lot.

<10 years old 0.27
10-19 years old 0.18
20+ years old 0.15



Attached House (Medium Density)
Characterized by individual units that share a common wall, with each unit on its own lot. Examples include townhomes and rowhouses.

<10 years old 0.27
10-19 years old 0.18
20+ years old 0.15



Attached House (High Density)
Characterized by individual units that share a common wall. Many high-density attached houses include shared open space amenities in backyards or courtyards. Examples include duplexes, triplexes and units with shared courtyards.

0.18

CORRIDOR APARTMENTS



Plex
A dwelling having apartments with separate entrances to six or more units. This includes two-story houses having a complete apartment on each floor and side-by-side apartments on a single lot that share a common wall.

0.18



Corridor Apartment
A four-story residential apartment building, typically with one on-street entrance and internal entrances to individual units.

0.03



Neighborhood Mixed Use
A four-story residential apartment building with commercial uses on the ground floor.

0.03



Single Room Occupancy Unit (SRO)
A studio apartment that does not have its own washing, laundry and kitchen facilities. Examples include affordable housing projects, assisted living facilities and college dormitories.

0.00

MID- TO HIGH-RISE APARTMENTS



Mid-Rise Mixed Use (Small Units)
A six- to ten-story building with ground floor office or retail uses. Allocated units of this type tend to be predominantly studios and one-bedroom units and tend to have smaller units.

0.02



Mid-Rise Mixed Use (Large Units)
A six- to ten-story building with ground floor office or retail uses. Typical units are larger, one- to four-bedroom units, and have a smaller number of studio units as part of the overall mix.

0.11



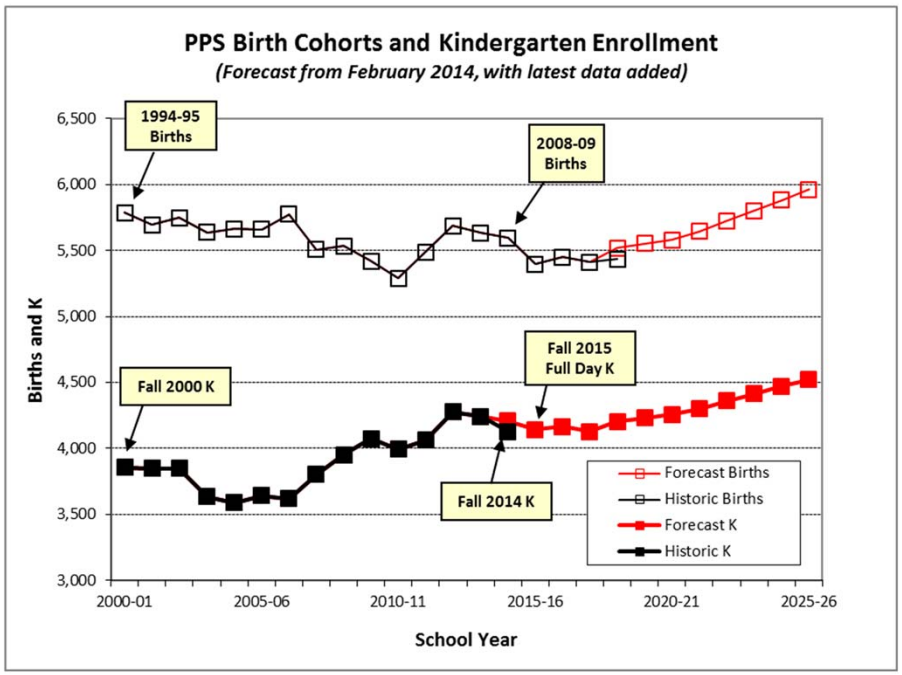
High-Rise Tower
A 10+ story building containing residential apartments or condominium units. In addition to spectacular views, most high rises offer their residents a full range of amenities. Building features may include 24-hour concierge service, swimming pools, spas, saunas, tennis courts, exercise areas, party rooms and guest suites.

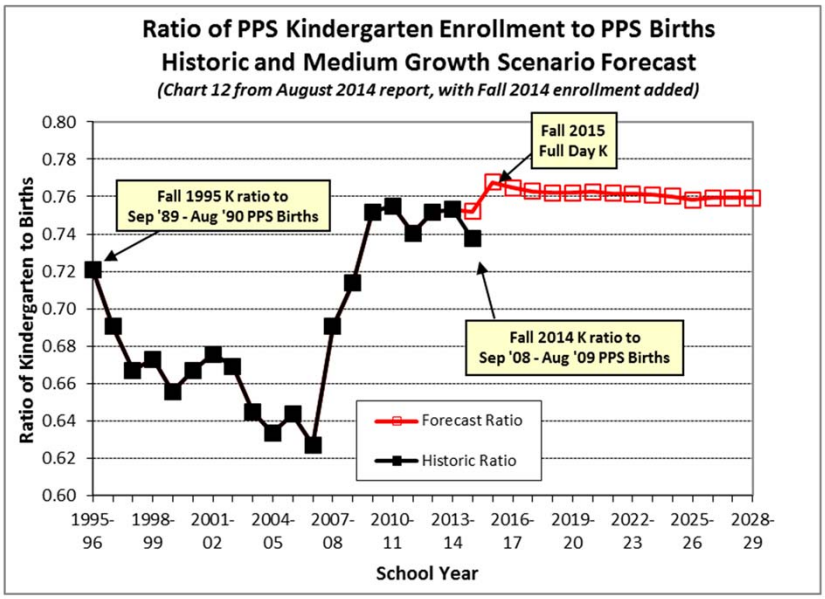
0.02

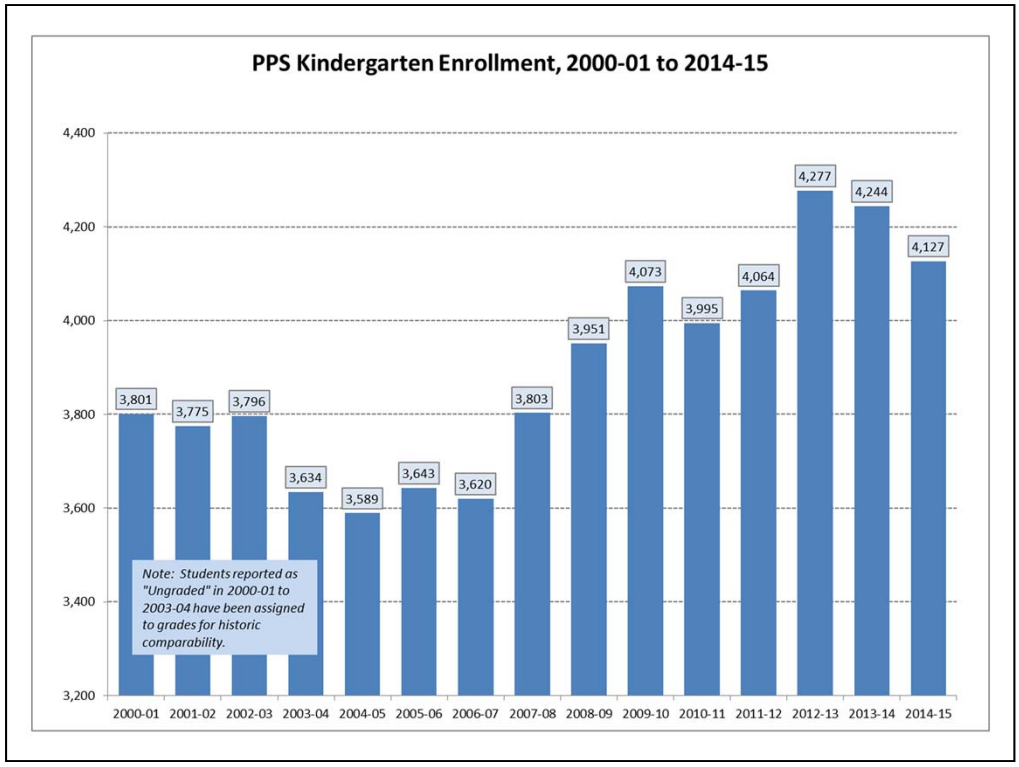
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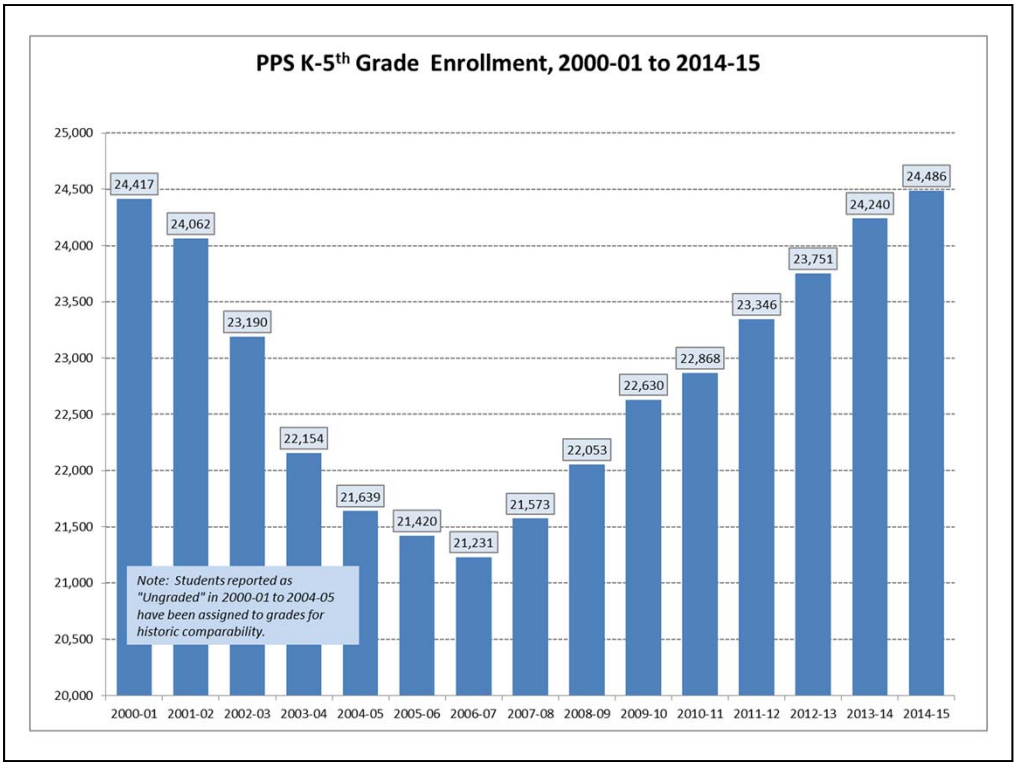
Enrollment Trends

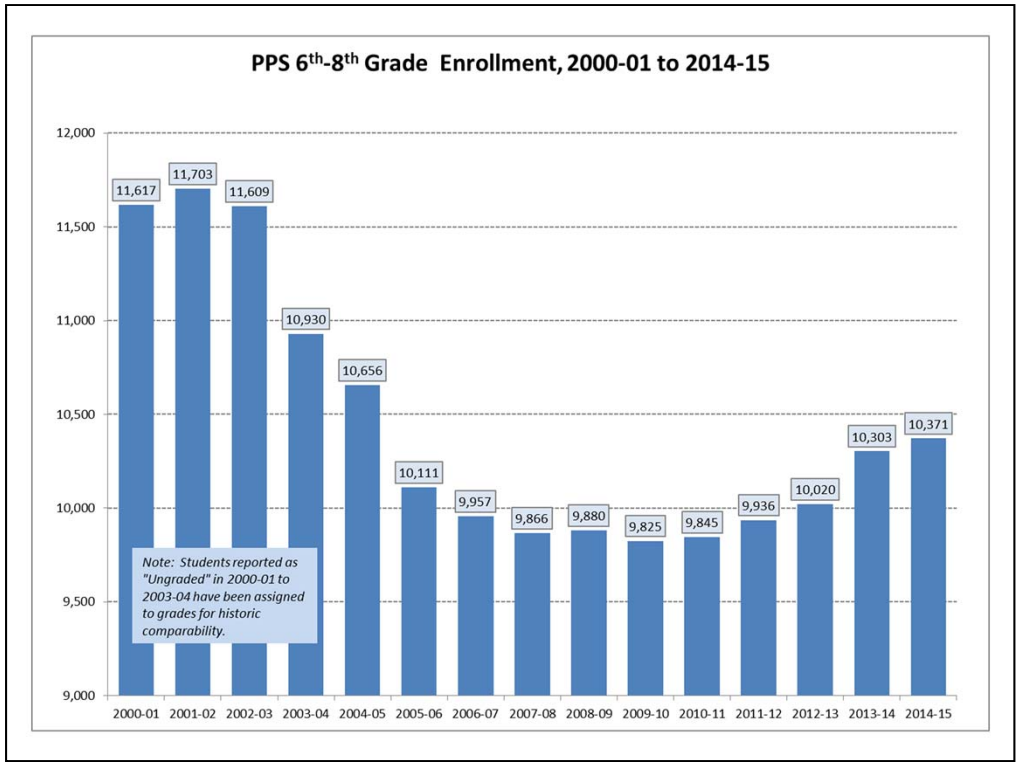
- Kindergarten/birth trend
- School level (K-5th, 6th-8th, 9th-12th)
- Historic grade progression rates
- Trends by cluster

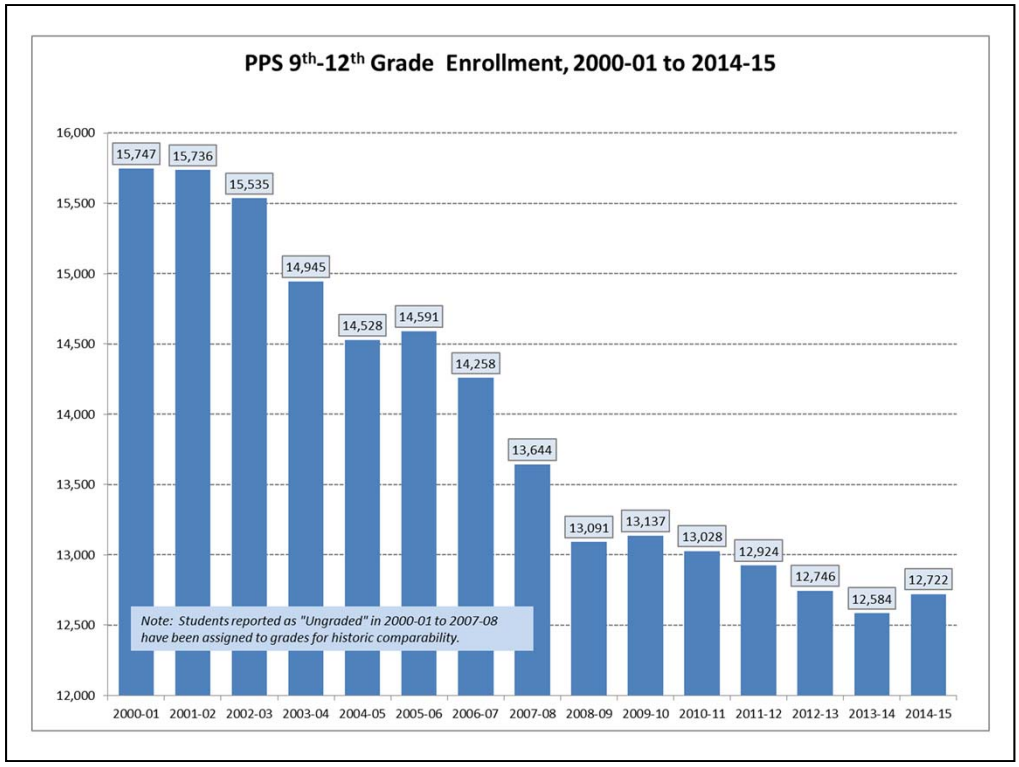

PPS Birth Cohorts and Kindergarten Enrollment
(Forecast from February 2014, with latest data added)
1994-95 Births
2008-09 Births
Fall 2000 K
Fall 2015 Full Day K
Fall 2014 K
Forecast Births
Historic Births
Forecast K
Historic K
Births and K
6,500
6,000
5,500
5,000
4,500
4,000
3,500
3,000
2000-01
2005-06
2010-11
2015-16
2020-21
2025-26
School Year

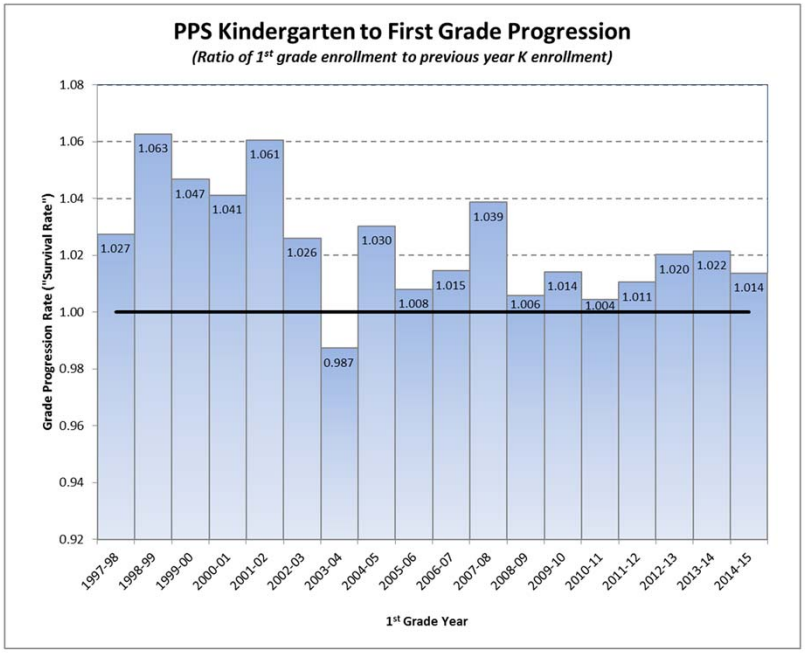

Ratio of PPS Kindergarten Enrollment to PPS Births
Historic and Medium Growth Scenario Forecast
(Chart 12 from August 2014 report, with Fall 2014 enrollment added)
Fall 2015
Full Day K
Fall 1995 K ratio to
Sep '89 - Aug '90 PPS Births
Fall 2014 K ratio to
Sep '08 - Aug '09 PPS Births
Forecast Ratio
Historic Ratio
Ratio of Kindergarten to Births
0.80
0.78
0.76
0.74
0.72
0.70
0.68
0.66
0.64
0.62
0.60
1995-96
1998-99
2001-02
2004-05
2007-08
2010-11
2013-14
2016-17
2019-20
2022-23
2025-26
2028-29
School Year

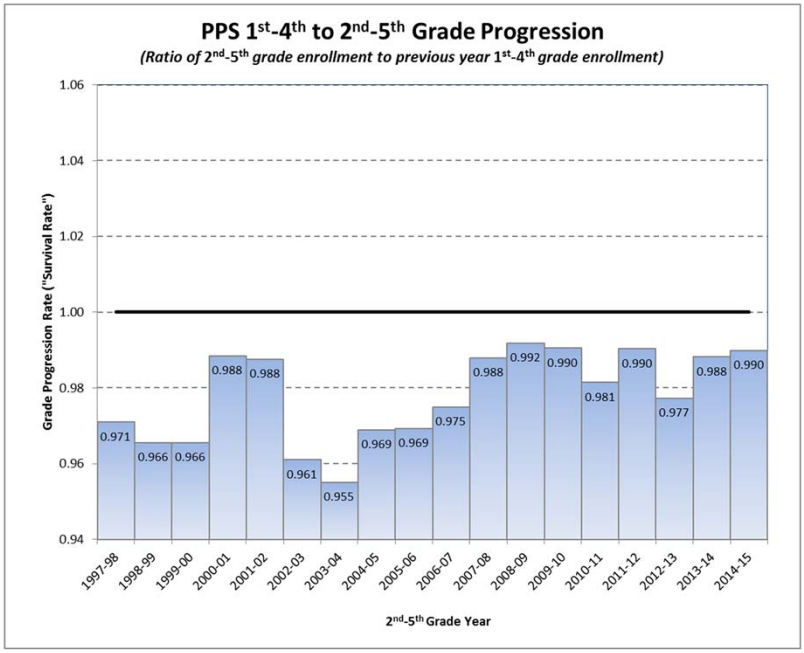

PPS Kindergarten Enrollment, 2000-01 to 2014-15
4,400
4,200
4,000
3,800
3,600
3,400
3,200
3,801
3,775
3,796
3,634
3,589
3,643
3,620
3,803
3,951
4,073
3,995
4,064
4,277
4,244
4,127
2000-01
2001-02
2002-03
2003-04
2004-05
2005-06
2006-07
2007-08
2008-09
2009-10
2010-11
2011-12
2012-13
2013-14
2014-15
Note: Students reported as "Ungraded" in 2000-01 to 2003-04 have been assigned to grades for historic comparability.

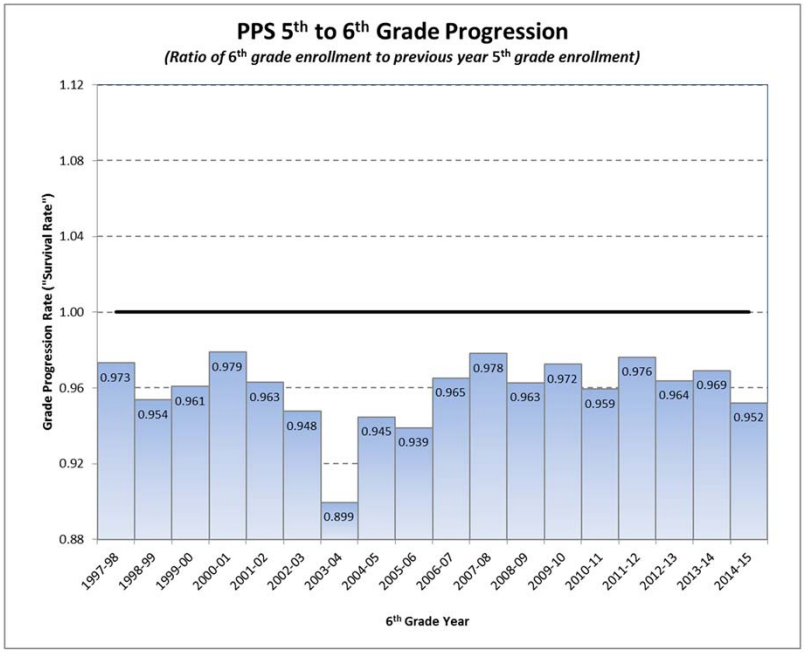

PPS K-5th Grade Enrollment, 2000-01 to 2014-15
25,000
24,500
24,000
23,500
23,000
22,500
22,000
21,500
21,000
20,500
20,000
24,417
24,062
23,190
22,154
21,639
21,420
21,231
21,573
22,053
22,630
22,868
23,346
23,751
24,240
24,486
Note: Students reported as "Ungraded" in 2000-01 to 2004-05 have been assigned to grades for historic comparability.
2000-01
2001-02
2002-03
2003-04
2004-05
2005-06
2006-07
2007-08
2008-09
2009-10
2010-11
2011-12
2012-13
2013-14
2014-15

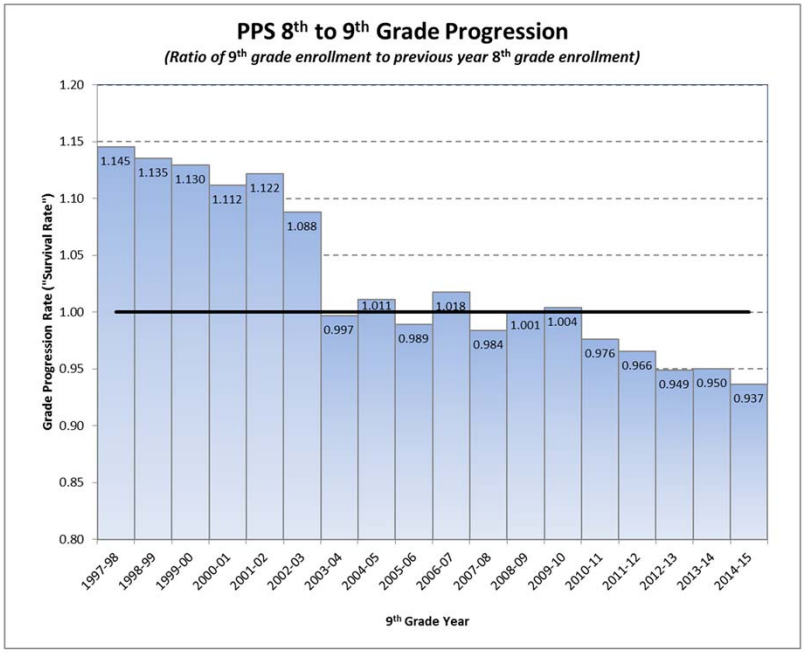

PPS 6th-8th Grade Enrollment, 2000-01 to 2014-15
12,000
11,500
11,000
10,500
10,000
9,500
9,000
11,617
11,703
11,609
10,930
10,656
10,111
9,957
9,866
9,880
9,825
9,845
9,936
10,020
10,303
10,371
2000-01
2001-02
2002-03
2003-04
2004-05
2005-06
2006-07
2007-08
2008-09
2009-10
2010-11
2011-12
2012-13
2013-14
2014-15
Note: Students reported as "Ungraded" in 2000-01 to 2003-04 have been assigned to grades for historic comparability.

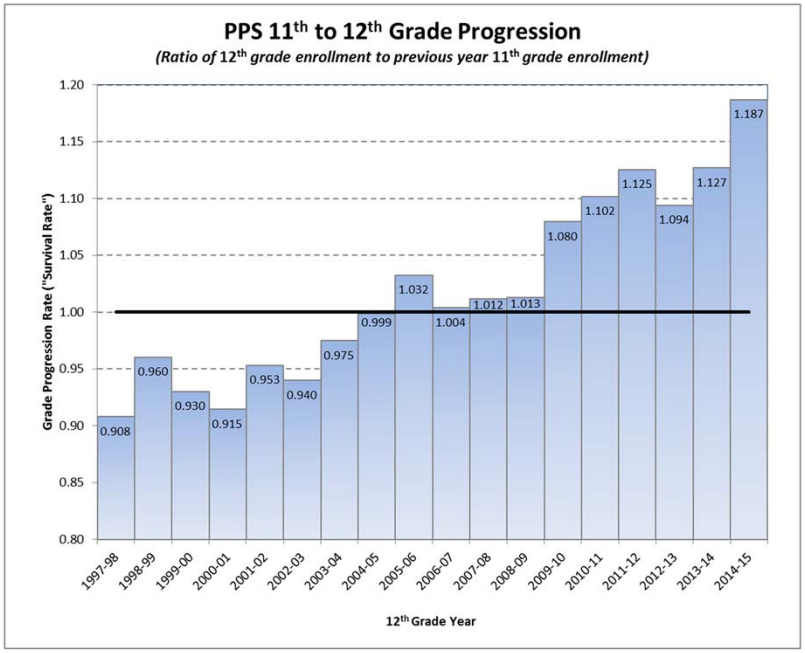

PPS 9th-12th Grade Enrollment, 2000-01 to 2014-15
16,000
15,500
15,000
14,500
14,000
13,500
13,000
12,500
12,000
15,747
15,736
15,535
14,945
14,528
14,591
14,258
13,644
13,091
13,137
13,028
12,924
12,746
12,584
12,722
2000-01
2001-02
2002-03
2003-04
2004-05
2005-06
2006-07
2007-08
2008-09
2009-10
2010-11
2011-12
2012-13
2013-14
2014-15
Note: Students reported as "Ungraded" in 2000-01 to 2007-08 have been assigned to grades for historic comparability.

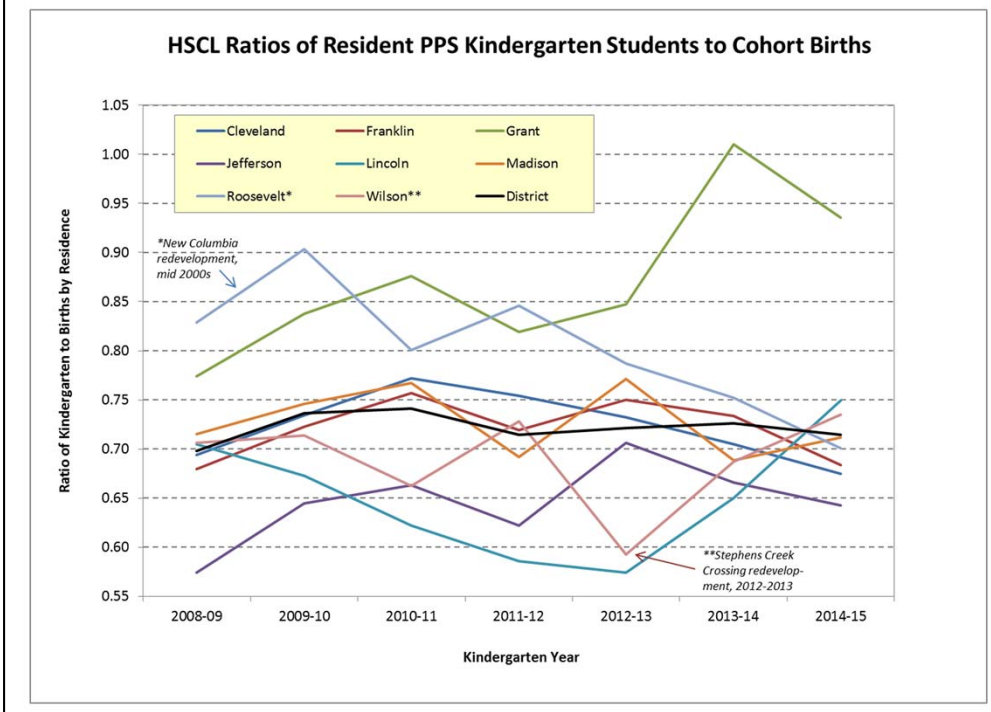

PPS Kindergarten to First Grade Progression
(Ratio of 1st grade enrollment to previous year K enrollment)
Grade Progression Rate ("Survival Rate")
1.08
1.06
1.04
1.02
1.00
0.98
0.96
0.94
0.92
1.027
1.063
1.047
1.041
1.061
1.026
0.987
1.030
1.008
1.015
1.039
1.006
1.014
1.004
1.011
1.020
1.022
1.014
1997-98
1998-99
1999-00
2000-01
2001-02
2002-03
2003-04
2004-05
2005-06
2006-07
2007-08
2008-09
2009-10
2010-11
2011-12
2012-13
2013-14
2014-15
1st Grade Year


PPS 1st-4th to 2nd-5th Grade Progression
(Ratio of 2nd-5th grade enrollment to previous year 1st-4th grade enrollment)
Grade Progression Rate ("Survival Rate")
1.06
1.04
1.02
1.00
0.98
0.96
0.94
0.971
0.966
0.966
0.988
0.988
0.961
0.955
0.969
0.969
0.975
0.988
0.992
0.990
0.981
0.990
0.977
0.988
0.990
1997-98
1998-99
1999-00
2000-01
2001-02
2002-03
2003-04
2004-05
2005-06
2006-07
2007-08
2008-09
2009-10
2010-11
2011-12
2012-13
2013-14
2014-15
2nd-5th Grade Year


PPS 5th to 6th Grade Progression
(Ratio of 6th grade enrollment to previous year 5th grade enrollment)
Grade Progression Rate ("Survival Rate")
1.12
1.08
1.04
1.00
0.96
0.92
0.88
0.973
0.954
0.961
0.979
0.963
0.948
0.899
0.945
0.939
0.965
0.978
0.963
0.972
0.959
0.976
0.964
0.969
0.952
1997-98
1998-99
1999-00
2000-01
2001-02
2002-03
2003-04
2004-05
2005-06
2006-07
2007-08
2008-09
2009-10
2010-11
2011-12
2012-13
2013-14
2014-15
6th Grade Year


PPS 8th to 9th Grade Progression
(Ratio of 9th grade enrollment to previous year 8th grade enrollment)
Grade Progression Rate ("Survival Rate")
1.20
1.15
1.10
1.05
1.00
0.95
0.90
0.85
0.80
1.145
1.135
1.130
1.112
1.122
1.088
0.997
1.011
0.989
1.018
0.984
1.001
1.004
0.976
0.966
0.949
0.950
0.937
1997-98
1998-99
1999-00
2000-01
2001-02
2002-03
2003-04
2004-05
2005-06
2006-07
2007-08
2008-09
2009-10
2010-11
2011-12
2012-13
2013-14
2014-15
9th Grade Year


PPS 11th to 12th Grade Progression
(Ratio of 12th grade enrollment to previous year 11th grade enrollment)
Grade Progression Rate ("Survival Rate")
1.20
1.15
1.10
1.05
1.00
0.95
0.90
0.85
0.80
0.908
0.960
0.930
0.915
0.953
0.940
0.975
0.999
1.032
1.004
1.012
1.013
1.080
1.102
1.125
1.094
1.127
1.187
1997-98
1998-99
1999-00
2000-01
2001-02
2002-03
2003-04
2004-05
2005-06
2006-07
2007-08
2008-09
2009-10
2010-11
2011-12
2012-13
2013-14
2014-15
12th Grade Year


HSCL Ratios of Resident PPS Kindergarten Students to Cohort Births
Ratio of Kindergarten to Births by Residence
1.05
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
0.60
0.55
Cleveland
Franklin
Grant
Jefferson
Lincoln
Madison
Roosevelt*
Wilson**
District
*New Columbia redevelopment, mid 2000s
**Stephens Creek Crossing redevelop-ment, 2012-2013
2008-09
2009-10
2010-11
2011-12
2012-13
2013-14
2014-15
Kindergarten Year

K-5th Grade Enrollment Growth by Resident High School Cluster

Sorted by Numeric Growth

High School Cluster	K-5th Residents		6 Year Change	
	2008-09	2014-15	Number	Percent
Cleveland	3,045	3,593	548	*18%*
Grant	2,282	2,736	454	*20%*
Wilson	2,178	2,494	316	*15%*
Lincoln	2,041	2,312	271	*13%*
Jefferson	2,901	3,139	238	*8%*
Franklin	3,869	4,102	233	*6%*
Madison	2,834	3,003	169	*6%*
Out of District	450	560	110	*24%*
Roosevelt	2,453	2,547	94	*4%*
PPS Total	**22,053**	**24,486**	**2,433**	***11%***

Forecast Accuracy: District-wide K-12 Total

School Year	Actual Enroll.[1]	K-12 Enrollment Forecasts by Base Year[2]									
		'04-'05	'05-'06	'06-'07	'07-'08	'08-'09	'09-'10	'10-'11	'11-'12	'12-'13	'13-'14
2004-05	46,823										
2005-06	46,122	45,875									
2006-07	45,446	45,304	45,404								
2007-08	45,083	44,754	44,711	44,833							
2008-09	45,024	44,229	43,968	44,200	44,729						
2009-10	45,592	43,753	43,361	43,613	44,534	45,046					
2010-11	45,741	43,429	42,852	43,024	44,406	45,092	45,653				
2011-12	46,206	43,252	42,596	42,693	44,357	45,288	45,993	45,979			
2012-13	46,517	43,350	42,656	42,508	44,611	45,696	46,588	46,451	46,661		
2013-14	47,127	43,350	42,666	42,659	44,651	45,886	46,979	46,766	46,901	46,980	
2014-15	47,579	43,470	42,688	42,693	44,789	46,226	47,420	47,325	47,268	47,544	47,617

School Year	Percentage Error in Enrollment Forecasts by Base Year[2]									
	'04-'05	'05-'06	'06-'07	'07-'08	'08-'09	'09-'10	'10-'11	'11-'12	'12-'13	'13-'14
2004-05										
2005-06	*-0.5%*									
2006-07	*-0.3%*	*-0.1%*								
2007-08	*-0.7%*	*-0.8%*	*-0.6%*							
2008-09	*-1.8%*	*-2.3%*	*-1.8%*	*-0.7%*						
2009-10	*-4.0%*	*-4.9%*	*-4.3%*	*-2.3%*	*-1.2%*					
2010-11	*-5.1%*	*-6.3%*	*-5.9%*	*-2.9%*	*-1.4%*	*-0.2%*				
2011-12	*-6.4%*	*-7.8%*	*-7.6%*	*-4.0%*	*-2.0%*	*-0.5%*	*-0.5%*			
2012-13	*-6.8%*	*-8.3%*	*-8.6%*	*-4.1%*	*-1.8%*	*0.2%*	*-0.1%*	*0.3%*		
2013-14	*-8.0%*	*-9.5%*	*-9.5%*	*-5.3%*	*-2.6%*	*-0.3%*	*-0.8%*	*-0.5%*	*-0.3%*	
2014-15	*-8.6%*	*-10.3%*	*-10.3%*	*-5.9%*	*-2.8%*	*-0.3%*	*-0.5%*	*-0.7%*	*-0.1%*	*0.1%*

1. Includes ungraded, excludes pre-kindergarten.

2. Previous reports included either one or three alternative forecast series. Forecasts presented in this table are those characterized as "Medium" when more than one alternative was prepared.

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Forecast Accuracy: District-wide by Grade Level

Grade	2014-15 Actual Enroll.	2014-15 Enrollment Forecasts by Base Year*							
		2013-14 (1 yr.)		2012-13 (2 yr.)		2011-12 (3 yr.)		2010-11 (4 yr.)	
		Fcst.	Error	Fcst.	Error	Fcst.	Error	Fcst.	Error
K	4,127	4,209	2.0%	4,211	2.0%	4,073	-1.3%	4,029	-2.4%
1	4,302	4,297	-0.1%	4,341	0.9%	4,184	-2.7%	4,141	-3.7%
2	4,287	4,304	0.4%	4,281	-0.1%	4,094	-4.5%	4,110	-4.1%
3	4,041	4,038	-0.1%	4,007	-0.8%	4,001	-1.0%	3,996	-1.1%
4	3,864	3,819	-1.2%	3,805	-1.5%	3,890	0.7%	3,881	0.4%
5	3,865	3,868	0.1%	3,794	-1.8%	3,886	0.5%	3,871	0.2%
6	3,594	3,667	2.0%	3,642	1.3%	3,678	2.3%	3,600	0.2%
7	3,428	3,514	2.5%	3,499	2.1%	3,509	2.4%	3,448	0.6%
8	3,349	3,378	0.9%	3,378	0.9%	3,396	1.4%	3,415	2.0%
9	3,137	3,211	2.4%	3,217	2.6%	3,258	3.9%	3,285	4.7%
10	3,090	3,082	-0.3%	3,172	2.7%	3,181	2.9%	3,257	5.4%
11	2,946	2,926	-0.7%	2,955	0.3%	3,000	1.8%	3,111	5.6%
12	3,549	3,304	-6.9%	3,242	-8.7%	3,118	-12.1%	3,181	-10.4%
Total	**47,579**	**47,617**	0.1%	**47,544**	-0.1%	**47,268**	-0.7%	**47,325**	-0.5%
Mean Absolute Pct. Error			**1.5%**		**2.0%**		**2.9%**		**3.1%**

*Note: Medium Growth Scenarios

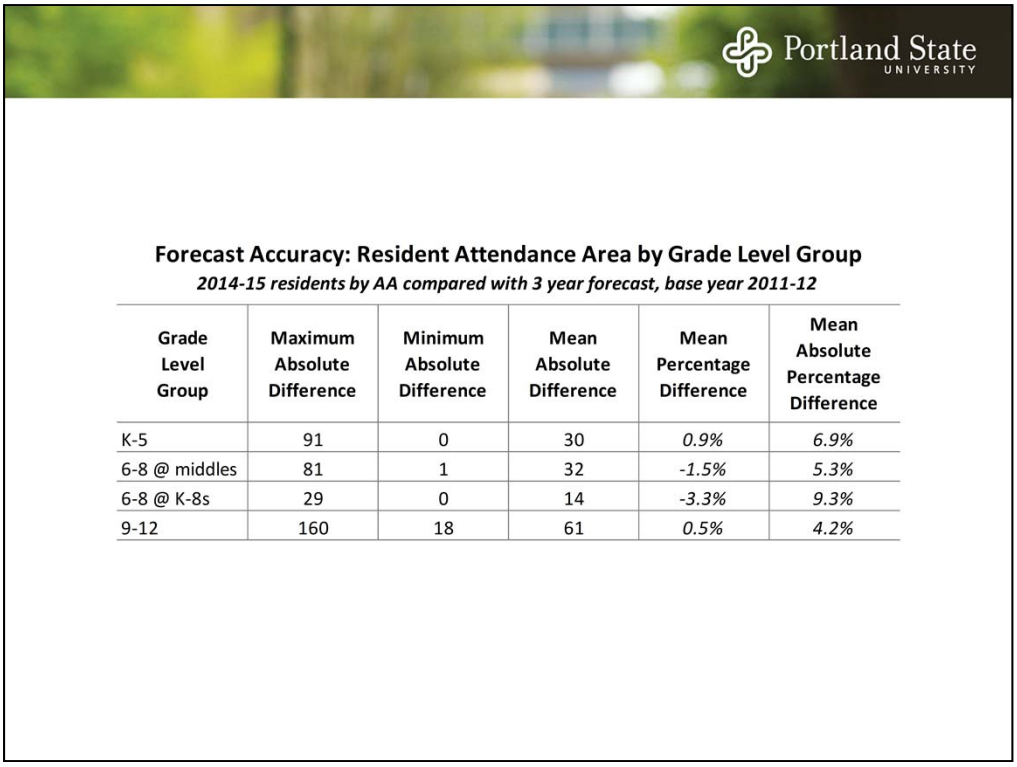
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Forecast Accuracy: Resident High School Cluster by Grade Level Group

2014-15 enrollment by cluster compared with 3 year forecast, base year 2011-12

Grade Level Group	Maximum Absolute Difference	Minimum Absolute Difference	Mean Absolute Difference	Mean Percentage Difference	Mean Absolute Percentage Difference
K-5	174	0	67	*1.6%*	*2.4%*
6-8	98	4	46	*-2.5%*	*4.1%*
9-12	160	19	61	*0.7%*	*4.0%*
K-12	406	11	109	*0.5%*	*2.1%*

Mean percentage difference shows bias. K-5 and 9-12 enrollments in 2014-15 were generally higher than forecast three years earlier, while 6-8 enrollments in 2014-15 were generally lower than forecast. All other measures are absolute.



Forecast Accuracy: Resident Attendance Area by Grade Level Group

2014-15 residents by AA compared with 3 year forecast, base year 2011-12

Grade Level Group	Maximum Absolute Difference	Minimum Absolute Difference	Mean Absolute Difference	Mean Percentage Difference	Mean Absolute Percentage Difference
K-5	91	0	30	*0.9%*	*6.9%*
6-8 @ middles	81	1	32	*-1.5%*	*5.3%*
6-8 @ K-8s	29	0	14	*-3.3%*	*9.3%*
9-12	160	18	61	*0.5%*	*4.2%*

Mean percentage difference shows bias. K-5 and 9-12 enrollments in 2014-15 were generally higher than forecast three years earlier, while 6-8 enrollments in 2014-15 were generally lower than forecast. All other measures are absolute.

Because of the size difference in 6th-8th grade enrollment between middle schools and K-8 schools, the K-8 numeric differences are smaller, while percentage differences are larger.

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Forecast Accuracy: Individual Schools by Type

2014-15 enrollment compared with 3 year forecast, base year 2011-12

School Type	Maximum Absolute Difference	Minimum Absolute Difference	Mean Absolute Difference	Mean Percentage Difference	Mean Absolute Percentage Difference
K-5/K-8*	89	0	33	*0.5%*	*7.5%*
Middle	53	3	25	*-1.4%*	*5.0%*
High**	144	7	59	*2.3%*	*4.9%*
Non-NH***	21	1	7	*-1.7%*	*1.8%*

**Excluding Chief Joseph, Duniway, Llewelyn, and Ockley Green, for which future changes were not known.*

***Including Benson.*

****Excluding ACCESS, for which future expansion was not known.*

Introduction to Enrollment Forecasts

Portland Public Schools

January 22, 2015



Charles Rynerson
Population Research Center
College of Urban and Public Affairs

Portland State UNIVERSITY